

PROBLEMS SUBMIT STATUS STANDINGS CUSTOM TEST

E. Sausage Maximization

time limit per test: 2 seconds
memory limit per test: 256 megabytes

The Bitlandians are quite weird people. They have their own problems and their own solutions. They have their own thoughts and their own beliefs, they have their own values and their own merits. They have their own dishes and their own sausages!

In Bitland a sausage is an array of integers! A sausage's deliciousness is equal to the bitwise excluding OR (the *xor* operation) of all integers in that sausage.

One day, when Mr. Bitkoch (the local cook) was going to close his BitRestaurant, BitHaval and BitAryo, the most famous citizens of Bitland, entered the restaurant and each ordered a sausage.

But Mr. Bitkoch had only one sausage left. So he decided to cut a prefix (several, may be zero, first array elements) of the sausage and give it to BitHaval and a postfix (several, may be zero, last array elements) of the sausage and give it to BitAryo. Note that one or both pieces of the sausage can be empty. Of course, the cut pieces mustn't intersect (no array element can occur in both pieces).

The pleasure of BitHaval and BitAryo is equal to the bitwise XOR of their sausages' deliciousness. An empty sausage's deliciousness equals zero.

Find a way to cut a piece of sausage for BitHaval and BitAryo that maximizes the pleasure of these worthy citizens.

Input

The first line contains an integer n ($1 \leq n \leq 10^5$).

The next line contains n integers $a_1, a_2, ..., a_n$ ($0 \leq a_i \leq 10^{12}$) — Mr. Bitkoch's sausage.

Please, do not use the `%lld` specifier to read or write 64-bit integers in C++. It is preferred to use the `cin`, `cout` streams or the `%I64d` specifier.

Output

Print a single integer — the maximum pleasure BitHaval and BitAryo can get from the dinner.

Examples

input	Copy
2 1 2	
output	Copy
3	

input	Copy
3 1 2 3	
output	Copy
3	

input	Copy
2 1000 1000	
output	Copy
1000	

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Round 173 (Div. 2)

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
282827380	Sep/24/2024 21:51	Accepted
282827041	Sep/24/2024 21:48	Wrong answer on test 33
282607417	Sep/23/2024 13:53	Accepted
282605975	Sep/23/2024 13:44	Wrong answer on test 33
282535788	Sep/23/2024 00:01	Wrong answer on test 8
282535745	Sep/23/2024 00:01	Wrong answer on test 8
282535619	Sep/22/2024 23:59	Wrong answer on test 8
282535595	Sep/22/2024 23:58	Compilation error
282535323	Sep/22/2024 23:55	Wrong answer on test 8

→ **Problem tags**

[bitmasks](#) [data structures](#) [trees](#) [*2200](#)

No tag edit access

→ **Contest materials**

- [Announcement](#) ✕
- [Round #173 — Editorial \(en\)](#) ✕

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